

ARTIFICIAL INTELLIGENCE SELF-LEARNING CLASSICAL KHON DANCE



Thanadon Poomkosum¹, Nathamon Thiengcharoen¹, Supachok Ruensom¹, Anun Wongsang¹, Nattapon Buaurai¹, Somchoke Ruengittinun²

¹ Suankularb Wittayalai Rangsit School, Pathum Thani 12120, Thailand
² Department of Computer Science, Faculty of Science, Kasetsart University, Bangkok 10900, Thailand

Introduction

- Khon is a traditional Thai masked dance drama and an important part of Thailand’s cultural heritage.
- Mastering Khon requires years of continuous training, and the limited number of Khon institutions and specialists in Thailand can make self-directed learning challenging.
- People's desire to learn about and interact with Khon has grown, which promotes appreciation and aids in the preservation of this significant cultural legacy.



Khon: The Crown Jewel of Thai Performance Arts
<https://thailandfoundation.or.th/khon-the-crown-jewel-of-thai-performance-arts/>

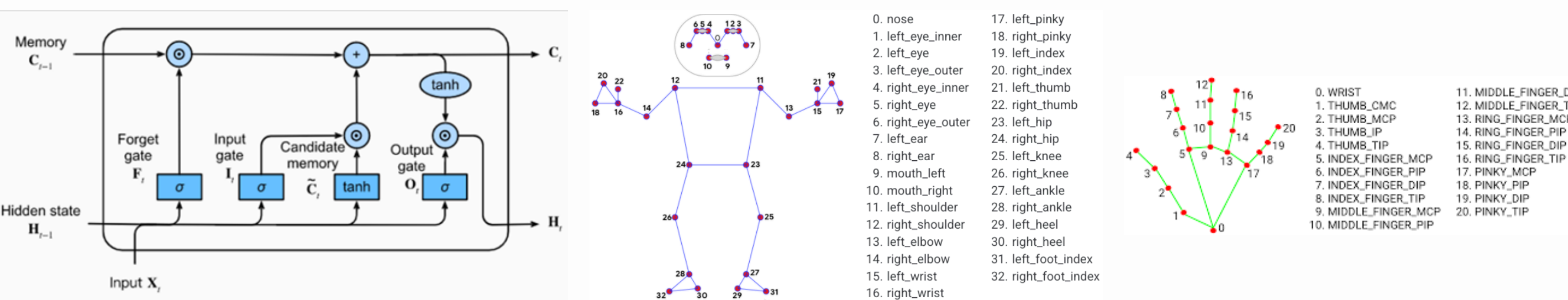
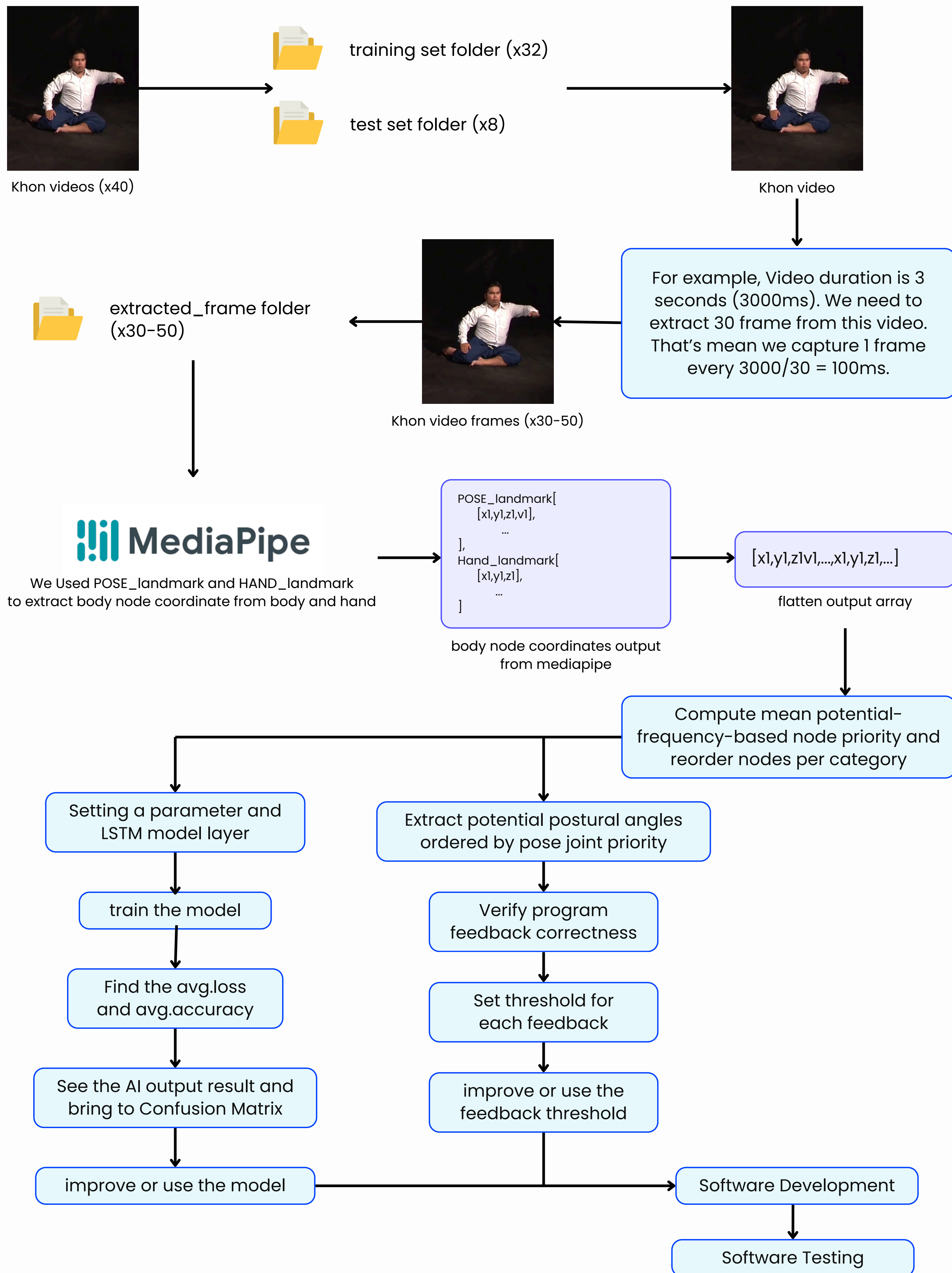


A sacred khon initiation
<https://www.nationthailand.com/life/30358739>

Objectives

- To create an artificial intelligence-driven virtual expert to assist in Khon dance training and practice.
- To create a web platform to make Khon dance learning accessible to the general public.
- To support interactive and self-paced learning for learners of all levels.

Methodology



Results and Discussions

We tested from a sample group of 20 students from the Khon and Thai Classical Dance Club of Suankularb Wittayalai Rangsit School people who had previously trained in Khon dancing and 20 students from the Thai dance class of Suankularb Wittayalai Rangsit School who were beginning to train, totaling 40 people. Data was collected on the average accuracy of users who practiced each Khon dance move three times. The average was found when the Khon expert assessed that the user had done everything correctly.

Model	Mean loss during training	Accuracy	F1 Score
Model 1	0.00472	0.837204	0.840297
Model 2	0.02213	0.792031	0.801762
Model 3	0.09948	0.720918	0.789302

Table I. Training Loss, Accuracy, and F1 Score of Each Model

posture	mean Accuracy (Already trained) [%]	mean Accuracy (Never trained) [%]	Mean Accuracy [%]
1. Tob Khao	96.82	84.48	90.65
2. Ten Sao	97.24	81.57	89.41
3. Thong Sa-eo	95.97	81.62	88.80

TABLE II. Top 3 Mean Accuracy Scores of Trained and Untrained Students

The following table shows the average scores of 40 students performing Khon dance postures before and after using the self-practice Khon application for one semester in Thai classical dance class (Table III).

Posture Category	Before		After		Improvement (Δ Score)	Students Improved [%]
	Average Score	Standard Deviation	Average Score	Standard Deviation		
fundamental motions	79.03	13.52	94.80	5.82	15.77	100
dance vocabulary	57.81	15.34	85.23	6.13	27.42	100
sign language	39.18	16.27	70.63	7.93	31.45	100

Table III. Student Scores Ststistic for Khon Dance Postures Before and After Using the Self-Practice Khon Application

Conclusions

- Students’ performance in Khon dance postures improved after using the self-practice Khon application.
- Compared to the previous study, scores improved across all postures.
- The application is effective in supporting learning and skill development.
- Future work: Apply the tool in larger classes or other dance contexts.

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